

Tinnaphop Sonsang

111/1 Moo 9, Nong Kratum Sub-district, Thap Than District, Uthai Thani

Email: tinnaphop.moss@gmail.com | Mobile: 064-948-1740 (please schedule via email before calling)

LinkedIn: [Tinnaphop Sonsang | LinkedIn](#) | GitHub: github.com/badgaitintinagain, github.com/badgaitintin

A Computer Engineering and AI graduate with hands-on experience building end-to-end AI pipelines from data engineering and model training to real-world deployment. Skilled in computer vision, data engineering, and full-stack web development, with a focus on making AI systems production-ready.

EDUCATION

Walailak University

B.Eng. Computer Engineering and Artificial Intelligence

GPAX: 3.65 (First-class honors)

Nakhorn Si Thammarat

June 2022 – 2026

Graduation: 7 May 2026

KEY SKILLS

End-to-End AI Pipeline Design

- Built multi-stage computer vision pipelines combining object detection, depth estimation, and pose estimation for real-world deployment
- Designed ensemble classification frameworks with Dynamic Entropy-based Weighting for improved prediction reliability
- Deployed YOLO-based WBC detection model achieving mAP@50 of 0.955

Data Engineering & Unsupervised Learning

- Processed and extracted 11+ audio features from 600K+ Spotify tracks using scalable data pipelines
- Applied K-Means, PCA, UMAP, t-SNE for dimensionality reduction, dataset deduplication, and image filtering

Full-Stack Web Development

- Frontend: Astro, Next.js, React, TypeScript, Tailwind CSS
- Backend: PostgreSQL (Supabase), SQLite, Drizzle ORM
- Integrated real-time AI inference endpoints into production web applications

Tools & Productivity

- Figma, Adobe Photoshop, Canva, Microsoft Office Suite

WORK EXPERIENCE

Intern Developer (Co-op/Internship) | ThaiDotRun, Bangkok

Sept 2025 – Apr 2026

Automated Runner Highlight Extraction System

- Designed an automated video clipping from running event pipeline using Object Detection Model (RT-DETR) and BoTSORT for subject tracking.
- Implemented a custom spatial weighting algorithm to prioritize subjects closest to the camera center, and integrated OCR to extract runner bib numbers. This dynamically replaced track IDs to determine precise start and end frames for automated highlight generation.
- Achieved an average success rate of approximately 60% for automated highlight video generation, establishing a solid baseline for initial deployment and future optimization.

Robustness Shoe Detection & Classification Pipeline

- Designed and developed an end-to-end deep learning pipeline for detecting and classifying brands from complex images and video footage.
- Applied unsupervised learning for dataset deduplication to eliminate redundant training frames. Utilized Grounding DINO to detect and localize runners, extracting shoe crops for an ensemble learning pipeline trained on the deduplicated dataset. Additionally, implemented depth-mapping and keypoint detection to assess image quality and enhance overall prediction reliability.
- Successfully integrated an ensemble classification framework, providing a comprehensive foundation for automated quality control and data re-collection.

PROJECTS

Madonna Discography Analysis

March 2026 - Present

- Developed an end-to-end music analytics system using unsupervised learning to identify hidden patterns within Madonna's 198-track discography, processed from a dataset of 600,000+ Spotify songs.
- Applied K-Means clustering, optimized via the Elbow Method to determine the ideal K-value, and utilized t-SNE to visualize and analyze high-dimensional relationships among the tracks.
- Achieved highly accurate track categorization by successfully identifying 4 distinct "Sonic Personas":

Energetic & Danceable (e.g., Girl Gone Wild, Get Together), Acoustic & Mellow (e.g., Love Profusion, Fever), Bold & Uplifting (e.g., Don't Cry For Me Argentina, Frozen), and Smooth & Sophisticated (e.g., Holiday, 4 Minutes).

Sitedotmoss

December 2025 – Present

- Developed a product-level, full-stack interactive portfolio demonstrating advanced state management, full-stack integration, and highly interactive UX design.
- Architected a highly responsive frontend utilizing Astro, React, TypeScript, and Tailwind CSS, paired with a backend powered by SQLite and Drizzle ORM. Seamlessly integrated Hugging Face Spaces to handle real-time AI inference.
- Designed a cohesive retro-futuristic UI/UX that blends past and future aesthetics. The interface incorporates Y2K, Vectordelia, and Frutiger Aero elements with glassmorphism and pastel tones, drawing structural inspiration from top-down cityscapes, architectural glass panels, and the Windows Phone tile system.

Capstone Project (Thesis)

August 2024 – August 2025

Detection of White Blood Cells in Raptors

- Developed a YOLO-based object detection model trained on a custom dataset of 1,200 images to accurately identify six classes of white blood cells in raptors.
- Architected a full-stack web application utilizing Next.js, React, and PostgreSQL (via Supabase) to deploy the model, enabling users to upload image batches and receive seamless diagnostic predictions.
- Achieved a high mean Average Precision (mAP@50) of 0.955, and an mAP@50-95 of 0.742 across multiple IoU thresholds.